The New Keynesian Model

Advanced Macroeconomics

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Motivation

Evidence that output drops when money supply contracts:

- eg France 1724. Government redenominates coins: money supply cut by 45%, prices adjust sluggishly, industrial production down by 30% (Velde 2009)

More recently, large amount of evidence that lower nominal interest rates increase output

- Long-standing view from central bankers
- Consistent with identified effects of monetary policy shocks from:
  - Structural VARs (eg Christiano, Eichenbaum, Evans 2005)
  - Narrative VARs (eg Romer and Romer 2004)
- Also, evidence from financial markets that monetary policy shocks affect real interest rates, including at fairly long horizons (eg Nakamura Steinsson 2016)

Basic Idea of the New Keynesian Model

In macro models so far, money is *neutral*: if the money stock is doubled (as a surprise), all the prices just double.

In the NK approach, a central bank will be more potent. Idea:

- Prices are sticky
- If the central banks issue money–e.g., by "helicopter drops" (lump-sum hand-outs in the population)–the price level would not change by the same proportion as the money increase
- Money is not neutral: there are real effects of monetary policy.

NK implementation is not through money handouts, however, but through manipulation of the nominal interest rate.

Roadmap

1. Evidence on Sticky Prices
2. Monopolistic Competition
3. The New Keynesian Model with Calvo prices
4. Sticky Wages

How often are prices changed? U.S. data

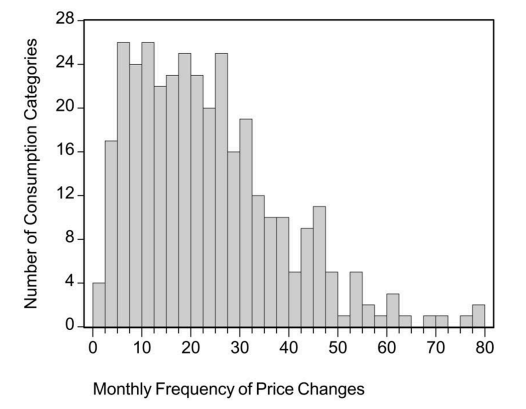

Figure 1: Source: Bils and Klenow 2004.

Within consumption category i, the fraction of goods for which the price is changed from one month to the next is x_i (x axis).

How often are prices changed? By year

TABLE 1
MONTHLY FREQUENCY OF PRICE CHANGES BY YEAR, 1995–2002

Year	Median Frequency (%)	Median Duration (Months)
1995	21.3	4.2
1996	20.8	4.3
1997	19.9	4.5
1998	21.2	4.2
1999	21.4	4.1
2000	21.7	4.1
2001–2	22.0	4.0

SOURCE.—U.S. Department of Labor, Commodities and Services Substitution Rate Table, various years.

NOTE.—2001–2 refers to the 15-month period from January 2001 through March 2002.

Figure 2: Source: Bils and Klenow 2004.

Is stickiness endogenous? It's important to know whether prices change in response to monetary policy!

U.S. vs. Europe: the euro area more sticky

Table 1 – Measures of price stickiness in the euro area and the US

(p.c. per month unless otherwise stated)

	Statistics	Euro area	US
CPI[1]	Frequency	15.1	24.8
	Average duration *(months)*	13.0	6.7
	Median duration *(months)*	10.6	4.6
PPI[2]	Frequency	20.0	n.a
Surveys[3]	Frequency	15.9	20.8
	Average duration *(months)*	10.8	8.3
NKPC[4]	Average durations *(months)*	13.5-19.2	7.2-8.4
Internet prices[5]	Frequency	95.5	94.7

[1] Dhyne *et al.* (2005) for the euro area, Bils and Klenow (2004) for the US. Euro area refers to the aggregate of Austria, Belgium, Finland, France, Germany, Italy, Luxembourg, the Netherlands, Portugal and Spain. [2] Vermeulen *et al.* (2005). Euro area corresponds to the aggregate of Belgium, Germany, Italy, Portugal and Spain. [3] Fabiani *et al.* (2005) for the euro area and Blinder *et al.* (1998) for the US. Euro area refers to the aggregate of Austria, Belgium, France, Germany, Italy, Luxembourg, the Netherlands, Portugal and Spain. Converted from original interval grouped figures [4] Gali *et al.* (2001, 2003). Estimates correspond to the GDP deflator and are converted from original quarterly figures. [5] Lünnemann and Wintr (2005). Euro area corresponds to the aggregate of Germany, France and Italy and are converted from original daily figures.

Figure 3: Source: Alvarez et al. 2005

Monopolistic Competition

- Stickiness requires “price setting”: perfect competition does not work.
- Monopolistic competition: tractable framework to embed IO considerations into GE
  - Widely used in other fields, e.g. trade
  - Idea: each monopolist faces downward-sloping demand curve
  - Their choice has infinitesimal effect on macro aggregates (really simplifies strategic considerations)
- Model: same backbone as before
  - Add continuum of goods (varieties), each produced by a monopolist
  - Remove capital accumulation to simplify
  - Consumption demand is largest component of Y, simplest to model
  - Investment demand is more complex, so we set aside in this class
  - Much exciting recent research enriches modeling of C and I

Preferences

- Static model first; varieties fixed. CES preferences:

$$u\left((c(i))_{i=0}^{1}\right) = \left(\int_0^1 c(i)^{1-\frac{1}{\epsilon}} di\right)^{\frac{\epsilon}{\epsilon-1}} \equiv C.$$

budget:

$$\int_0^1 p(i)c(i)di = y$$

Implication 1: a perfect price index

Good allocation problem:

$$\min_{(c(i))_{i=0}^{1}} \int_0^1 p(i)c(i)di \quad \text{s.t.} \quad \left(\int_0^1 c(i)^{1-\frac{1}{\epsilon}} di\right)^{\frac{\epsilon}{\epsilon-1}} \geq C$$

We obtain

$$p(i) = \lambda \left(\frac{c(i)}{C}\right)^{-\frac{1}{\epsilon}}$$

Solve for $c(i)$, use definition of C to get λ, and then find total spending:

$$\lambda = \left(\int_0^1 p(i)^{1-\epsilon} di\right)^{\frac{1}{1-\epsilon}} \equiv P \quad \text{and} \quad \int_0^1 p(i)c(i)di = PC$$

P is the price index, or price "per unit of C" (under optimal behavior).

Implication 2: the demand function

Plug in the solution for λ into the first-order condition:

$$c(i) = C\left(\frac{p(i)}{P}\right)^{-\epsilon}$$

- So constant price elasticity of demand. (One can replace C by y/P.)
- This makes the firm's problem easy (in a static setting).

The monopolistic competitor

- Assume linear technology in labor: $c(i) = An(i)$.
- Marginal cost of a unit of output is W/A.
- The firm i chooses $(c(i), p(i))$ to maximize

$$p(i)c(i) - \frac{W}{A}c(i) \quad \text{s.t.} \quad p(i) = P\left(\frac{c(i)}{C}\right)^{-\frac{1}{\epsilon}}$$

giving (if $\epsilon > 1$, otherwise infinite profits)

$$\frac{c(i)}{C} = \left(\frac{W}{A}\frac{1}{\left(1-\frac{1}{\epsilon}\right)P}\right)^{-\epsilon} \quad \text{and} \quad p(i) = \frac{\epsilon}{\epsilon - 1}\frac{W}{A}$$

- $\mu \equiv \epsilon/(\epsilon - 1) > 1$: constant percentage markup over marginal cost.

Monopolistic equilibrium and dynamic models

- Static model: all firms make the same choices: $P = \mu \frac{W}{A}$, or $W/P = A/\mu$.
- Normalize: $P = 1$. We obtain $W = A/\mu < A$. Optimal labor supply $\Rightarrow (C, h)$ from $u_c(C, h)\frac{A}{\mu} = u_h(C, h)$ and $C = Ah \equiv Y^n$: "natural" output.

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- Dynamic model flexible (real) prices: we assume (for simplicity) that there is no physical capital. Then each period is a repetition of the previous one. All firms still make the same choices and we see that $P = \mu \frac{W}{A}$ or $W/P = A/\mu$. Natural output Y_t^n varies if A_t varies.
- Real interest rate will be determined by a standard Euler equation. If A is constant the gross real interest rate will equal $1/\beta$.

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- Real interest rate will be determined by a standard Euler equation. If A is constant the gross real interest rate will equal $1/\beta$.
- Dynamic model, sticky (nominal) prices: now the firm problem is dynamic and there is a role for the central bank. P will, in general, vary over time.

Costs of changing prices

Two common assumptions used in applied work:

- Calvo prices (used below): a form of menu costs. Each period, a firm can reset with probability $1-\theta$ ("the firm is visited by the Calvo fairy"). I.e., cost of changing prices: either 0 or ∞ according to an exogenous Poisson process. (With general menu costs: choice.)
- Rotemberg pricing: there is a resource cost associated with going from price $p_{t-1}(i)$ to $p_t(i)$ equal to $g\left(p_t(i)/p_{t-1}(i)\right)$; g is quadratic and $g(1)=0$.

In case 1, firms all behave differently when the economy is subject to aggregate shocks; in case 2 , they all behave the same.

There are also models with other formulations (like random finite costs of changing prices: state-, not time-dependent settings).

Dynamic Model with Sticky Prices

- Let us generalize so that output of good i, $y(i)$, is $An(i)^{1-\alpha}$. Then
  - total costs are $W\left(\frac{y(i)}{A}\right)^{\frac{1}{1-\alpha}}$
  - and the marginal cost is increasing in output.
- We also allow A to be random: an aggregate TFP shock, as in RBC
- Firms now post prices in anticipation of costs conditions.
- They make a commitment to sell at that price.
  - These firms can't change price when information realizes in state s^t
  - If demand is higher than expected, so will their marginal cost, and hence their markup will be lower than optimal

The Calvo firm's problem

When it gets a price-reset opportunity, firm i chooses $P_t(i)$, and hence $Y_{t+s|t}(i)$ for all $s \geq 0$, to

$$\max \sum_{s=0}^{\infty} \theta^s E_t Q_{t,t+s} \left(Y_{t+s|t}(i) P_t(i) - W_{t+s} \left(\frac{Y_{t+s|t}(i)}{A_{t+s}} \right)^{\frac{1}{1-\alpha}} \right)$$

$$\text{subject to } Y_{t+s|t}(i) = \left(\frac{P_t(i)}{P_{t+s}} \right)^{-\epsilon} C_{t+s}$$

- $Q_{t,t+s}$ is the nominal discount factor, or the market price at time t of delivery of a unit of money at $t+s$.
- All firms that can reset do it the same way: $P_t(i) = P_t^*$.

Implications

- The FOC gives P_t^* as a function of $\{Q_{t,t+s}, P_{t+s}, W_{t+s}, C_{t+s}\}_{s=0}^{\infty}$ (forward-looking) and A_t . Conceptually, the reset price is related to marginal costs, current and future, relative to those of others, and to the overall price levels.
- The overall price level satisfies:

$$P_t^{1-\epsilon} = (1-\theta)\left(P_t^*\right)^{1-\epsilon} + \theta P_{t-1}^{1-\epsilon}$$

  Here, though many prices are preset, P is never a state variable since part of it is set in the current period.
- So inflation satisfies:

$$\Pi_t^{1-\epsilon} = \theta + (1-\theta)\left(\Pi_t^*\right)^{1-\epsilon}$$

  with $\Pi_t^* = \frac{P_t^*}{P_{t-1}}$ 'reset price inflation' (useful concept)

The consumer's problem in aggregate form

The representative household has standard preferences

$$E_0 \sum_{t=0}^{\infty} \beta^t u\left(C_t, N_t\right)$$

where C is the index of different varieties, and N is total labor $\left(\int_0^1 n(i)di\right)$. The budget in nominal terms is

$$P_t C_t + Q_t B_{t+1} = B_t + W_t N_t + \Pi_t \quad \forall t \geq 0$$

where Q_t is the price of nominally riskless one-period bonds B_{t+1} paying out 1 dollar at $t+1$. ($1/Q_t$ is the gross nominal interest rate between t and $t+1$.) Π_t is monopoly profits.

Optimality

Assume MaCurdy utility. Taking FOC, we have, for all t,

$$C_t^{-\sigma} = \beta E_t \left[\frac{1}{Q_t} \frac{1}{P_{t+1}} C_{t+1}^{-\sigma} \right] \quad \text{and} \quad C_t^{-\sigma} \frac{W_t}{P_t} = \psi N_t^{\frac{1}{\varphi}}$$

The (state-dependent) nominal discounting from t to $t+s$ is thus

$$Q_{t,t+s} = \beta^s \left(\frac{C_{t+s}}{C_t} \right)^{-\sigma} \cdot \frac{P_t}{P_{t+s}}$$

So we can substitute the Qs into the firm's FOC. We replace W_t from the consumer's intratemporal FOC; then N_t appears in the firm's FOC.

Market clearing

Market clearing reads

$$C_t = Y_t = \left(\int_0^1 y_t(i)^{1-\frac{1}{\epsilon}} di \right)^{\frac{\epsilon}{\epsilon-1}}$$

with $y_t(i) = A_t n_t(i)^{1-\alpha}$, and

$$N_t = \int_0^1 n_t(i) di$$

Price Dispersion and Aggregation

- In general, since firms are stuck at different prices and employ and produce differently, the distribution of hours worked and output matters for aggregates. . . messy!
- But we do have the $(1-\theta)\left(1,\theta,\theta^2,\ldots\right)$ structure, so we can obtain nice expressions, though with some serious footwork (omitted here, phew), for the aggregates.
- Aggregate output and hours satisfy:

$$Y_t = \frac{A_t}{\Delta_t} N_t^{1-\alpha}$$

- Price dispersion

$$\Delta_t \equiv \left[\int_{i=0}^{1} \left(\frac{p_{ti}}{P_t}\right)^{-\frac{\epsilon}{1-\alpha}} di\right]^{1-\alpha}$$

Solving further

One cannot solve the model in closed form.

One can derive two very simple equations: log-linearizations around a zero-inflation ($\Pi = 1$) and constant A steady state, where all firms are identical. The two equations are:

- the consumer's Euler equation: the "New-Keynesian IS curve";
- the firm's FOC, aggregated: the "forward-looking Phillips curve"

Linearizing the consumer's Euler equation

Let us derive the IS curve, in deterministic equilibrium:

$$Y_t^{-\sigma} = \beta \frac{1+i_t}{1+\pi_{t+1}} Y_{t+1}^{-\sigma} \Rightarrow \left(\frac{Y_t}{Y_t^n}\right)^{-\sigma} = \left(\frac{Y_{t+1}}{Y_{t+1}^n}\right)^{-\sigma} \frac{1+i_t}{1+\pi_{t+1}} \beta \left(\frac{Y_{t+1}^n}{Y_t^n}\right)^{-\sigma}$$

where Y^n is undistorted (RBC) output. With $Y/Y^n \equiv 1+\tilde{y}$, we have

$$(1+\tilde{y}_t)^{-\sigma} = (1+\tilde{y}_{t+1})^{-\sigma} \frac{1+i_t}{1+\pi_{t+1}} \frac{1}{1+r_t^n}$$

with r^n the undistorted real interest rate. Take logs: approximately,

$$\tilde{y}_t = \tilde{y}_{t+1} - \frac{1}{\sigma}\left(i_t - \pi_{t+1} - r_t^n\right)$$

"Demand" (the output gap, $\tilde{y}$) is stimulated by a lower i !

Linearizing the firm's FOC: the Phillips curve

Log-linearize around the zero-inflation steady state. The reset price satisfies:

$$p_t^* - p_{t-1} = (1-\beta\theta)\sum_{k=0}^{\infty}(\beta\theta)^k E_t\left\{\widehat{mc}_{t+k|t} + (p_{t+k} - p_{t-1})\right\}$$

where $\widehat{mc}_{t+k|t}$ is the log deviation of real marginal cost from steady state. Using the fact that an individual firm's marginal cost can be related to the economy-wide average:

$$\widehat{mc}_{t+k|t} = \widehat{mc}_{t+k} - \frac{\alpha\epsilon}{1-\alpha}(p_t^* - p_{t+k})$$

one can rewrite the reset price equation as a difference equation:

$$p_t^* - p_{t-1} = \beta\theta\, E_t\left\{p_{t+1}^* - p_t\right\} + (1-\beta\theta)\,\Theta\,\widehat{mc}_t + \pi_t$$

where $\Theta \equiv \frac{1-\alpha}{1-\alpha+\alpha\epsilon} \leq 1$.

The New Keynesian Phillips Curve

Combining with the log-linearized price index $\pi_t = (1-\theta)(p_t^* - p_{t-1})$ yields:

$$\boxed{\pi_t = \beta E_t \pi_{t+1} + \lambda \widehat{mc}_t}$$

where $\lambda \equiv \frac{(1-\theta)(1-\beta\theta)}{\theta}\Theta$. Using the fact that $\widehat{mc}_t$ is proportional to the output gap:

$$\widehat{mc}_t = \frac{\sigma + \varphi + \alpha}{1-\alpha}\tilde{y}_t$$

we obtain the NKPC in terms of the output gap:

$$\pi_t = \beta E_t \pi_{t+1} + \kappa \tilde{y}_t \qquad \text{where } \kappa \equiv \lambda \frac{\sigma + \varphi + \alpha}{1-\alpha}$$

Solving forward: $\pi_t = \kappa \sum_{k=0}^{\infty} \beta^k E_t \{\tilde{y}_{t+k}\}$. Inflation is high when firms expect output to remain above its natural level, because this signals high marginal costs.

Closing the Model: The Taylor Rule

The central bank sets the nominal interest rate, $1 + i_t \equiv 1/Q_t$, according to a rule. It is typically a "Taylor rule": raise the rate when inflation rises above trend, or when output goes above trend.

$$i_t = \rho + \phi_\pi \pi_t + \phi_y \tilde{y}_t + v_t$$

Key features:

- ρ is the steady-state real rate (recall: zero inflation)
- $\phi_\pi > 1$: necessary for a unique equilibrium, and for the cashless limit to work; note that if π goes up (by other forces), i goes up more, so that $i - \pi$ falls, and hence lowers demand
- $\phi_y > 0$: again, leaning against the wind
- v is a "monetary policy shock" (unforeseen)

The Taylor Principle and Determinacy

Substituting the Taylor rule into the IS and Phillips curves, the equilibrium is locally unique if and only if:

$$\kappa(\phi_\pi - 1) + (1-\beta)\phi_y > 0$$

- Intuition: suppose inflation permanently rises by $d\pi$. The Phillips curve implies $d\tilde{y} = \frac{1-\beta}{\kappa} d\pi$. The Taylor rule then adjusts i by:

  $$di = \left(\phi_\pi + \phi_y \frac{1-\beta}{\kappa}\right) d\pi$$

  Determinacy requires the term in parentheses to exceed 1: the real interest rate must eventually rise in the face of higher inflation, so that demand falls and the inflation increase is unsustainable.
- When $\phi_y = 0$: reduces to $\phi_\pi > 1$ (the nominal rate must rise more than one-for-one with inflation).
- This is the Taylor principle.

The full picture: the 3-equation system

The "three-equation Keynesian model" under uncertainty is:

$$\tilde{y}_t = E_t\tilde{y}_{t+1} - \frac{1}{\sigma}\left(i_t - E_t\pi_{t+1} - E_t r_t^n\right) \quad \text{(the NK IS curve)}$$

$$\pi_t = \beta E_t\pi_{t+1} + \kappa\tilde{y}_t \quad \text{(the Phillips curve)}$$

$$i_t = \rho + \phi_\pi\pi_t + \phi_y\tilde{y}_t + v_t \quad \text{(the Taylor rule)}$$

Note:

- the first equation is what we derived, plus $E_t s$
- the second equation is the firm's price-setting equation, elaborated on (price changes appear, not the level, because the price is not a state); $\tilde{y} > 0$ raises prices since higher output makes marginal costs rise.

Notice that a productivity shock appears indirectly: in r_t^n. A_t going up makes y_t^n rise and r_t^n fall, stimulating $\tilde{y}_t$. The Taylor rule partly fights it.

Expansionary Monetary Policy

$$i_t = \rho + \phi_\pi \pi_t + \phi_y \tilde{y}_t + v_t$$

- Joint determination of aggregate demand and inflation, given monetary policy
- Suppose $v_t < 0$ (stimulation). The real interest rate decreases.
- This raises aggregate demand via intertemporal substitution
- This raises marginal costs, and creates inflation
- If a firm can reset its price, it sets a higher one. But inflation will not fully respond, due to sticky prices.
- Output goes up since the higher demand stimulates wages (not sticky); also, profits fall. People work more: active labor supply channel.

Comments

Immensely popular model at central banks. It captures how central banks themselves view their role!

A couple of "issues":

- if only prices are sticky (as above) and there is no capital, the reason why output goes up when the fed lowers i is a bit phony: W/P goes up but with balanced-growth preferences, this doesn't make people work more, as the income effect cancels the substitution effect-what makes people work more is instead that profits fall (higher real wages), making workers poorer (a negative income effect)
- forward guidance "too" powerful: a lower i far out in the future raises y today as if the rate cut happened today; the IS curve is not discounted. And it raises inflation over the whole period, making the price level jump higher today the further out the change is planned.

Optimal Policy: the Approximate Problem

- We can compute a second-order approximation to welfare:

$$-\frac{1}{2}\mathbb{E}\left[\sum_{t=0}^{\infty}\beta^t\left\{\frac{\sigma+\varphi+\alpha}{1-\alpha}\left(y_t-y_t^e\right)^2+\frac{\epsilon}{\lambda}\pi_t^2\right\}\right]$$

  where $y_t^e \equiv \log Y_t^e$ is the log efficient output level and $\lambda \equiv \frac{(1-\theta)(1-\beta\theta)}{\theta}\Theta$
  - see Gali's pp 86-89 for details
- Optimal policy maximizes welfare subject to

$$\pi_t = \beta E_t \pi_{t+1} + \kappa \tilde{y}_t$$

- Once solution is computed, back out nominal interest rate using:

$$\tilde{y}_t = E_t\tilde{y}_{t+1} - \frac{1}{\sigma}\left(i_t - E_t\pi_{t+1} - r_t^n\right)$$

  so

$$i_t = r_t^n + E_t\pi_{t+1} + \sigma(E_t\tilde{y}_{t+1} - \tilde{y}_t)$$

The Divine Coincidence

- Consider the case where the government subsidizes production and taxes away profits, so $y_t^e = y_t^n$.
- Approximate welfare is then:

$$\max -\frac{1}{2}\mathbb{E}\left[\sum_{t=0}^{\infty}\beta^t\left\{\frac{\sigma+\varphi+\alpha}{1-\alpha}\left(\tilde{y}_t\right)^2+\frac{\epsilon}{\lambda}\pi_t^2\right\}\right]$$

  subject to

$$\pi_t = \beta E_t \pi_{t+1} + \kappa \tilde{y}_t$$

- Optimal solution is $\tilde{y}_t = \pi_t = 0$ at all t
- Divine coincidence result: inflation targeting is optimal, and goes hand-in-hand with output stabilization
  - Zero inflation avoids relative price distortions ($\Delta_t = 1$ always)
  - Can get the first best level of output at all times
  - Along this solution, the nominal interest rate $i_t = r_t^n$

Costs of Inflation

- Why don't people like inflation?
  - Menu costs, costs of searching for lowest price
  - Redistributes nominal debt from debtors to creditors (if unexpected)
  - Creates uncertainty/volatility, in addition to high average level
  - Creates confusion between relative price changes and price level changes
- Here: inefficient allocation of resources
  - captures some of these stories, nicely interpretable in terms of welfare
  - but maybe more of a parable for something else
- Outside of this model: why could $\pi > 0$ be optimal?
  - Risk of hitting the zero lower bound
  - Downward nominal wage rigidities ('greasing the wheels')

Frequency in euro area: wages (vs. prices)

Alternative to sticky prices is sticky wages: Workers are committed to work whatever firms ask for at W

Table 3 - Frequency of price and wage changes across sectors

(percentages)

	PRICES				
	daily to monthly	quarterly to half yearly	yearly	less frequently than once a year	no pattern
Total	**9.2**	**15.4**	**39.3**	**7.4**	**28.4**
Manufacturing	5.3	16.1	43.4	7.8	27.2
Construction	7.4	20.5	29.5	7.4	34.9
Trade	22.7	20.2	27.5	3.2	26.2
Market services	5.4	10.3	44.3	9.9	29.9
Financial services	15.1	19.0	22.9	5.3	35.9

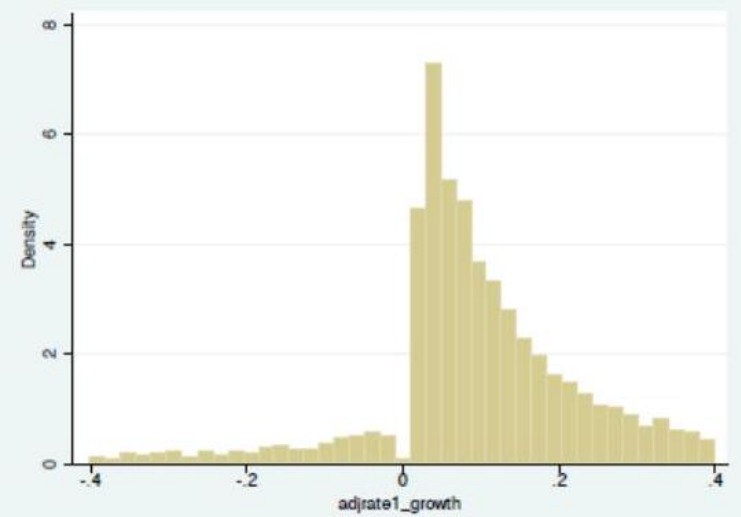
	WAGES (for any reason)			
	more frequently than once a year	yearly	less frequently than once a year	never/don't know
Total	**12.1**	**59.3**	**25.8**	**2.8**
Manufacturing	12.3	59.0	26.6	2.1
Construction	21.6	59.5	15.8	3.1
Trade	10.8	58.1	27.6	3.5
Market services	11.0	60.5	25.1	3.4
Financial services	16.0	58.4	24.0	1.5

Figures weighted by employment weight, rescaled excluding non-responses.

Fabiani et al. (2006): Wages are significantly stickier!

U.S.: the distribution of wage changes

Distribution of Non-Zero Wage Changes, Hourly Workers, 1996 Panel



Medium-scale NK models

Bells and whistles: Smets and Wouters (2003, 2007), Christiano, Eichenbaum, and Evans (2005).

Typical additional features:

- sticky wages
- wage and price indexation
- capacity utilization costs
- adjustment costs to investment
- consumption habits
- a bunch more shocks

Also, models estimated formally, typically with Bayesian methods.

Minnesota critique: Chari, Kehoe, and McGrattan (2009). Too many shocks and parameters

Remarks

- ”Large-scale” models also used at central banks.
- They guide, but do not steer, policy.
- Important open-economy versions too. (Exchange rates, terms of trade. . .)
- Models with unemployment exist, replacing labor supply channel.
- Many things missing—apparent after 2007-2009 recession:
- credit-market frictions
- housing
- asset prices
- banks
- heterogeneous agents (HANK)

Very much ongoing work, which has not entirely landed.

- The model should be viewed as a ”business-as-usual” framework. In crises/in response to large shocks, we may need different model features.

Appendix

Solving the Model: Monetary Policy Shock

Assume $v_t = \rho_v v_{t-1} + \varepsilon_t^v$ with $\rho_v \in [0, 1)$ and constant A (so $\hat{r}_t^n = 0$). Guess $\tilde{y}_t = \psi_{yv} v_t$, $\pi_t = \psi_{\pi v} v_t$. Substituting into IS and NKPC:

$$\tilde{y}_t = -(1 - \beta\rho_v)\Lambda_v v_t \qquad \pi_t = -\kappa\Lambda_v v_t$$

where $\Lambda_v \equiv \frac{1}{(1-\beta\rho_v)[\sigma(1-\rho_v)+\phi_y]+\kappa(\phi_\pi-\rho_v)} > 0$ under the Taylor principle.

- A contractionary shock ($v_t > 0$) lowers both output gap and inflation.
- The real rate rises: $\hat{r}_t = \sigma(1 - \rho_v)(1 - \beta\rho_v)\Lambda_v v_t > 0$.
- Since Y^n is unaffected, $\hat{y}_t = \tilde{y}_t$: output falls one-for-one with the gap.

Solving the Model: Technology Shock

Assume $a_t = \rho_a a_{t-1} + \varepsilon_t^a$ with $\rho_a \in [0,1)$ and $v_t = 0$. Then $\hat{r}_t^n = -\sigma\psi_{ya}^n(1-\rho_a)a_t$ where $\psi_{ya}^n = \frac{1+\varphi}{\sigma(1-\alpha)+\varphi+\alpha}$.

$$\tilde{y}_t = -(1-\beta\rho_a)\Lambda_a\,\sigma\psi_{ya}^n(1-\rho_a)\,a_t \qquad \pi_t = -\kappa\Lambda_a\,\sigma\psi_{ya}^n(1-\rho_a)\,a_t$$

- A positive technology shock lowers the output gap and inflation (as long as $\rho_a < 1$).
- Intuition: natural output rises, but sticky prices prevent the economy from fully adjusting. The central bank partially accommodates (lowers i), but not enough to close the gap.
- Employment can fall after a positive technology shock — the economy produces more per worker, but demand doesn't rise enough. This is consistent with the SVAR evidence (Galí 1999) and distinguishes the NK model from the RBC model.

Implementing the Optimal Policy

Along the optimal allocation, $i_t = r_t^n$. Can the central bank simply set $i_t = r_t^n$ as its rule?

- No: this rule leads to indeterminacy. The system under $i_t = r_t^n$ has one eigenvalue inside and one outside the unit circle, but both $\tilde{y}_t$ and π_t are non-predetermined $\Rightarrow$ multiple equilibria (Blanchard-Kahn).
- The optimal outcome $\tilde{y}_t = \pi_t = 0$ is *an* equilibrium, but *not the unique one*. Sunspot fluctuations can arise.
- Solution: use a rule with endogenous feedback, e.g. $i_t = r_t^n + \phi_\pi \pi_t + \phi_y \tilde{y}_t$, satisfying the Taylor principle. The "threat" of a strong response to deviations rules them out in equilibrium.
- Practical problem: r_t^n is unobservable — it depends on the true model, all parameter values, and all realized shocks. This motivates the study of "simple rules" that respond only to observable variables (e.g. $i_t = \rho + \phi_\pi \pi_t$).